DIRECT SUMMANDS OF AN M-PSEUDO-COHERENT OBJECT ARE M-PSEUDO-COHERENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the localisation and boundedness reduction, the displayed triangle and the cone lemma, the shift induction, and the backward descent to L.

Editorial note. The label lemma-summands-pseudo-coherent names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Triangles attached to a direct sum. Derived Categories, the lemma producing distinguished triangles from a split pair of objects. Supplies the triangle (K + L, K + L, L + L[1]) = (K,K,0) + (L,L,L + L[1]) used to isolate L.
- Cones of pseudo-coherent maps. Same chapter, the lemma stating that the cone on a map between m-pseudo-coherent objects is m-pseudo-coherent. Supplies the pseudo-coherence of L + L[1] from the triangle.

Source reference. Original lemma 05QS: sources/stacks-chow/tags.txt, source lines 7386–7386. Prerequisite pointer invoked at proof line(s) 12296; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 064R: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 12177–12177. Prerequisite pointer invoked at proof line(s) 12298; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $m \in \mathbf{Z}$. If $K \oplus L$ is m-pseudo-coherent (resp. pseudo-coherent) in $D(\mathcal{O})$ so are K and L.*

Proof. Assume that $K \oplus L$ is m-pseudo-coherent. Let U be an object of $\mathcal{C}$. After replacing U by the members of a covering we may assume $K \oplus L \in D^-(\mathcal{O}_U)$, hence $L \in D^-(\mathcal{O}_U)$. Note that there is a distinguished triangle

$$(K \oplus L, K \oplus L, L \oplus L[1]) = (K, K, 0) \oplus (L, L, L \oplus L[1])$$

see Derived Categories, Lemma 05QS. By Lemma 064R we see that $L \oplus L[1]$ is m-pseudo-coherent. Hence also $L[1] \oplus L[2]$ is m-pseudo-coherent. By induction

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$L[n] \oplus L[n+1]$ is m-pseudo-coherent. Since L is bounded above we see that $L[n]$ is m-pseudo-coherent for large n. Hence working backwards, using the distinguished triangles

$$(L[n], L[n] \oplus L[n-1], L[n-1])$$

we conclude that $L[n-1], L[n-2], \ldots, L$ are m-pseudo-coherent as desired. $\square$